

Bank Loan Analysis SQL Project Report

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Course: BBA (IT & System Management)

Abstract

This project designs a relational database for a bank loan management system using MySQL. It models customers, branches, employees, loan types, loans and payments while demonstrating practical SQL skills.

Introduction

Banks process thousands of loan applications. A structured database helps manage customer information, loan approvals, repayments and reporting accurately.

Problem Statement

Maintain customer, loan and payment information efficiently while supporting reporting and analysis.

Objectives

Create a normalized database, enforce relationships with primary and foreign keys, insert realistic data, perform analysis using SQL, and implement views, procedures and triggers.

Scope

Suitable for learning SQL, database design and analytics. Can be extended with EMI schedules, user authentication and audit logs.

Software Requirements

MySQL Workbench, GitHub, Microsoft Excel, Power BI (optional).

Database Design

Tables: Customers, Branches, Employees, LoanTypes, Loans and Payments. Relationships: Customers→Loans, LoanTypes→Loans, Branches→Employees, Branches→Loans, Employees→Loans, Loans→Payments.

Table Description

Customers stores customer details. Branches stores branch information. Employees stores bank staff. LoanTypes stores loan categories. Loans stores issued loans. Payments stores repayment transactions.

SQL Features

DDL: CREATE DATABASE/TABLE. DML: INSERT, UPDATE, DELETE. Constraints: PRIMARY KEY, FOREIGN KEY. Analysis: JOINS, GROUP BY, HAVING, Aggregate Functions, Subqueries, Views, Stored Procedures, Triggers.

Sample Business Questions

Total loans by branch, loan amount by type, active vs closed loans, customers with highest loan amounts, payment status summary, monthly loan trends.

Testing

Verified table creation, inserted sample records, executed joins, created view, procedure and trigger successfully.

Conclusion

The project demonstrates real-world relational database design and SQL analysis suitable for a portfolio and interviews.

Future Scope

Add EMI calculation, authentication, interest recalculation, dashboards and predictive analytics.